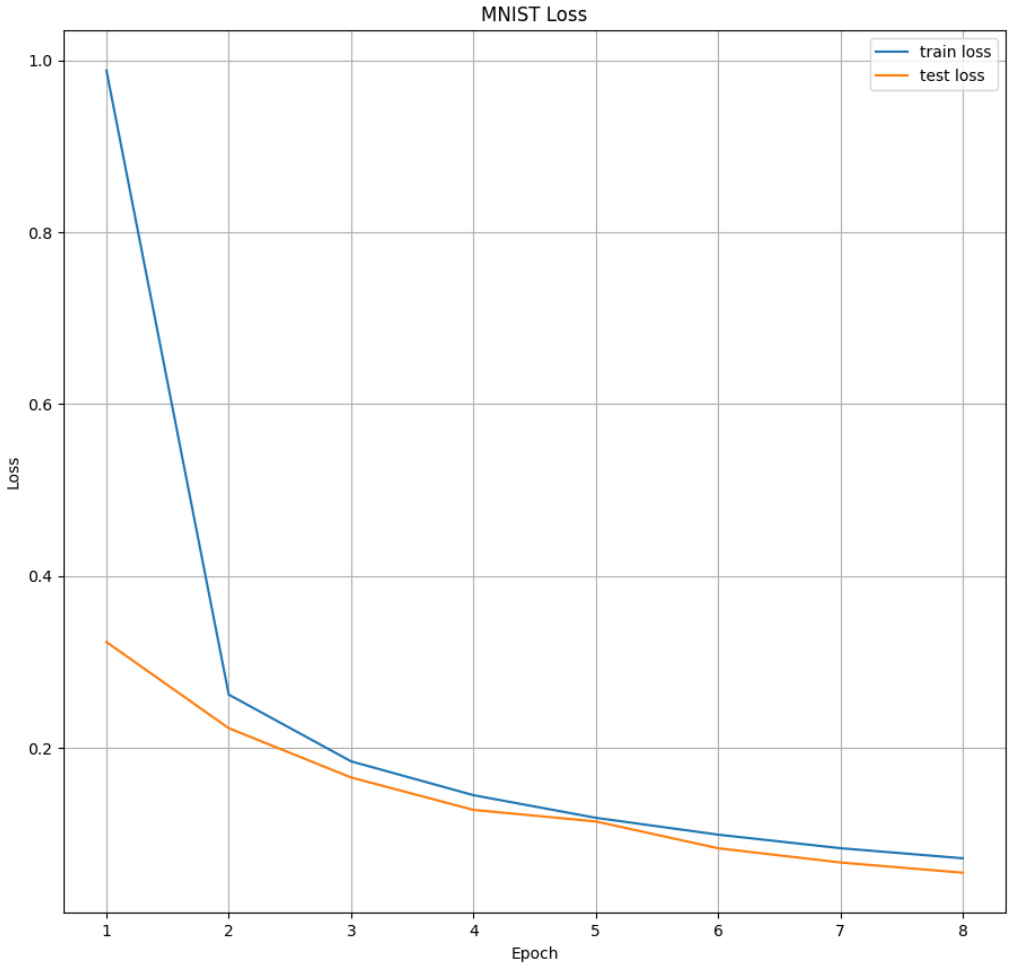
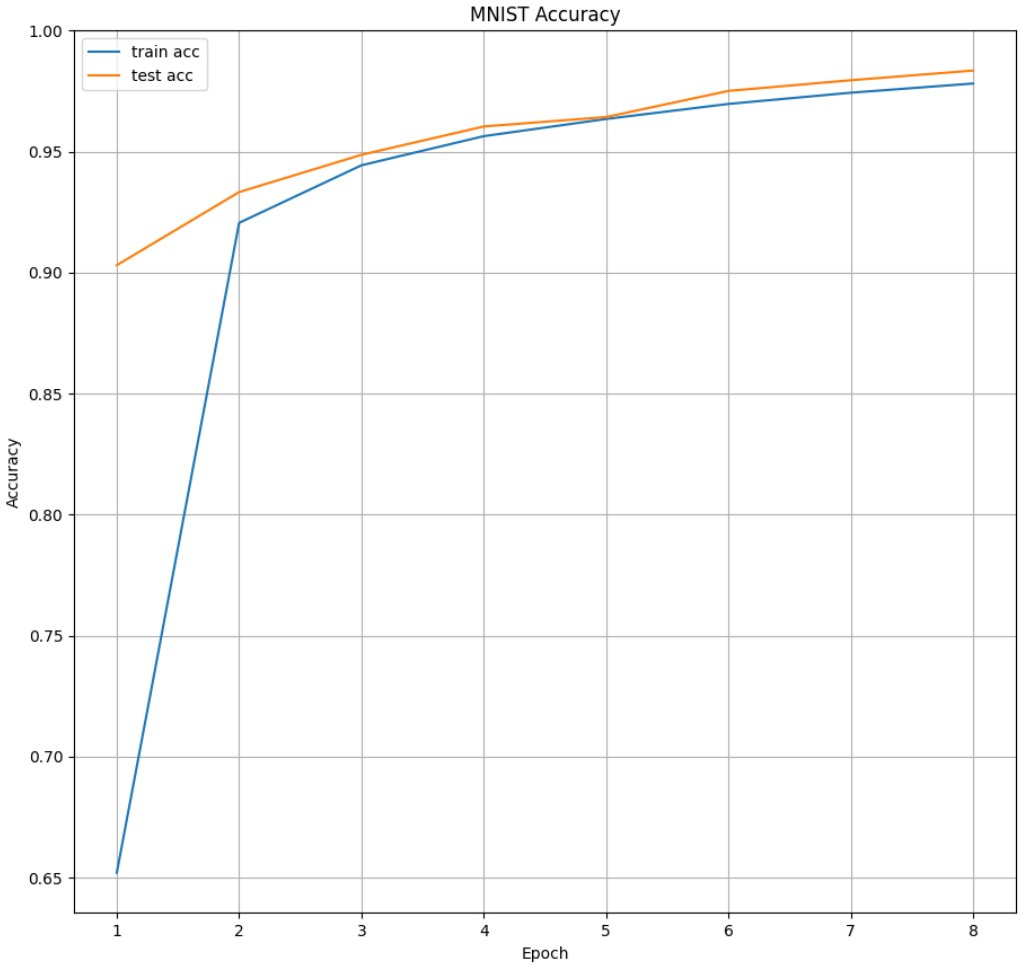
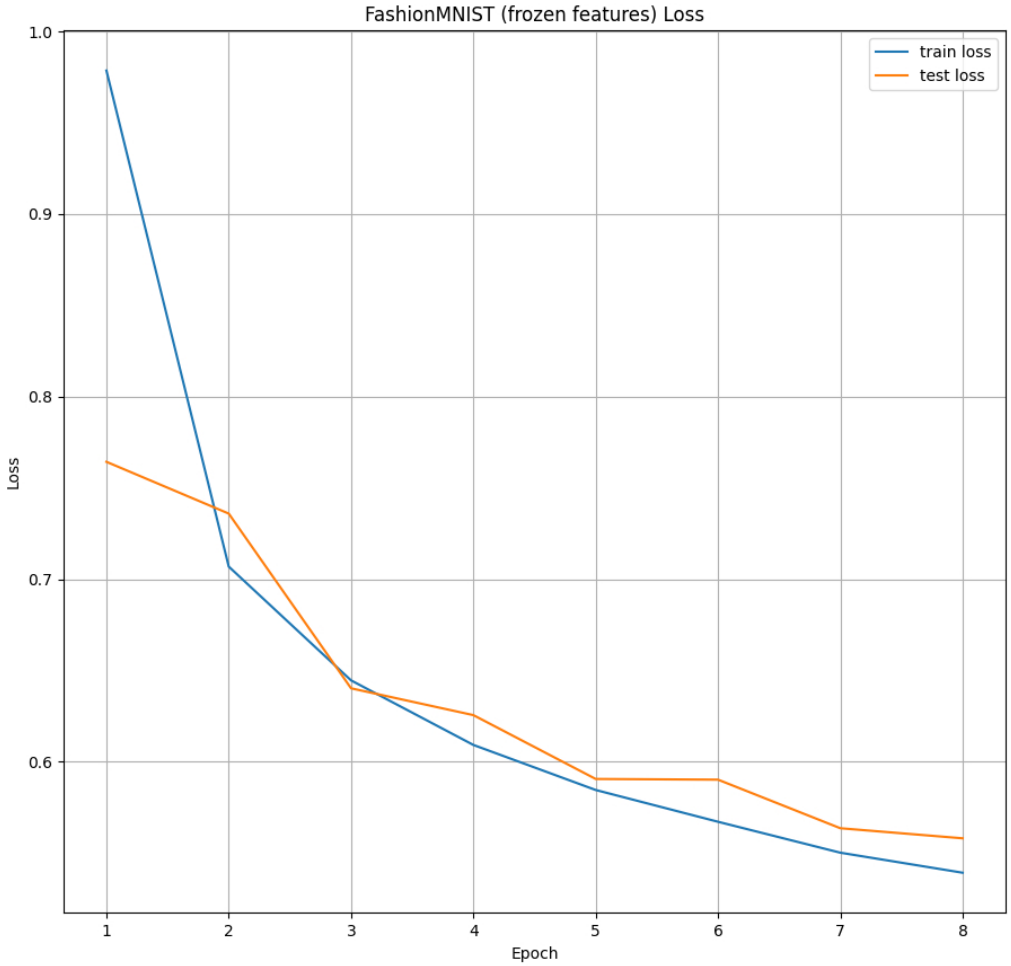
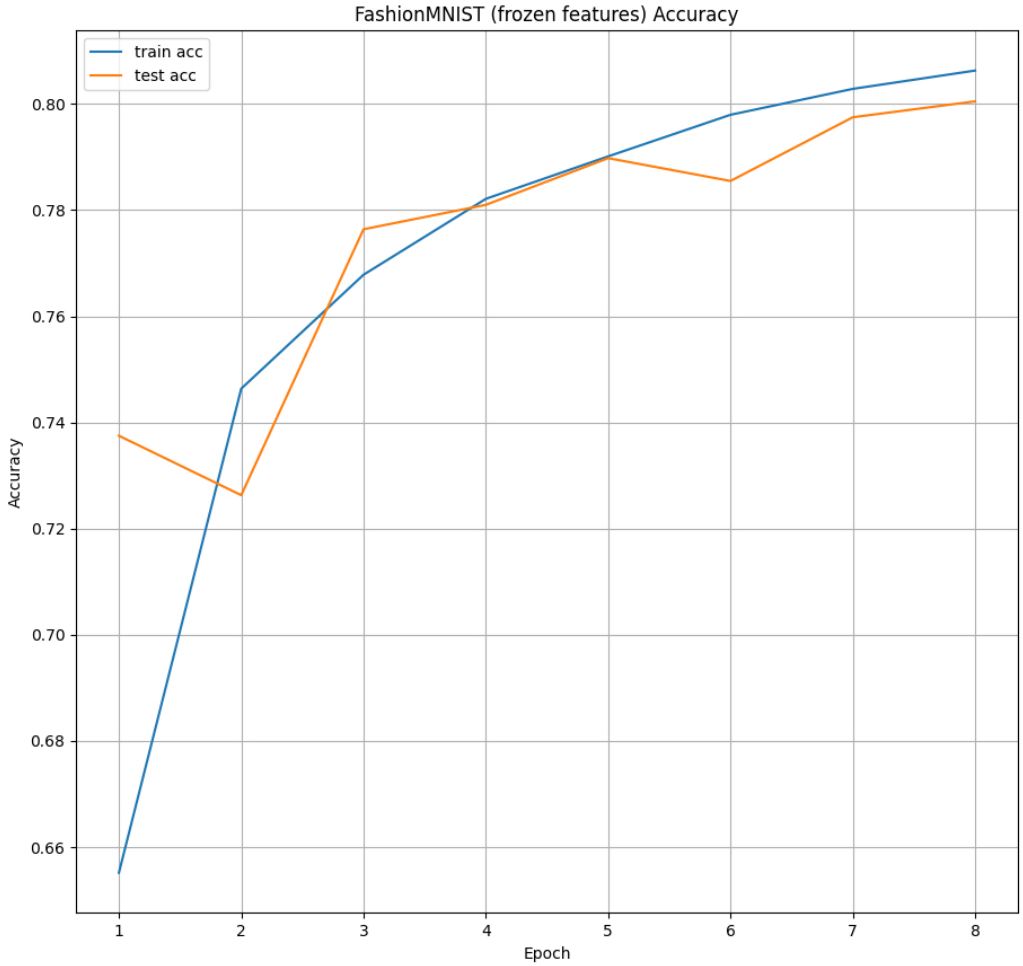
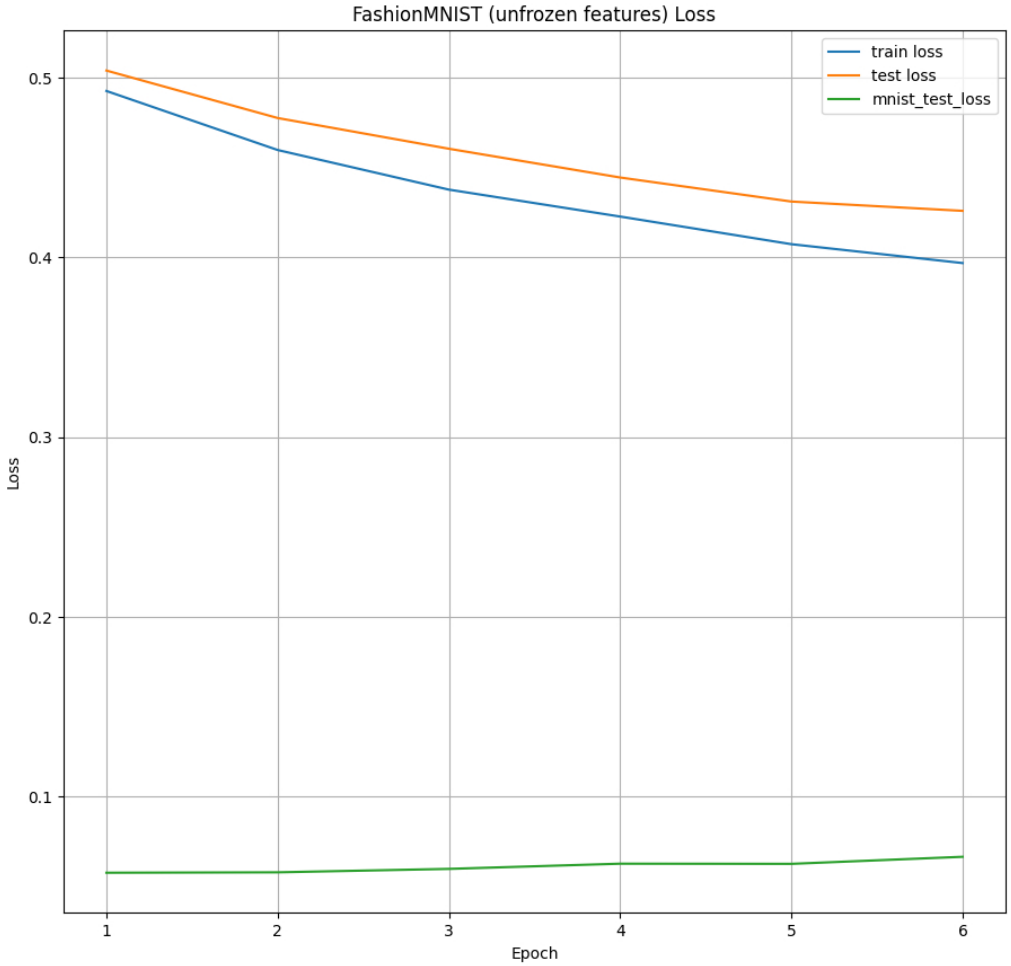
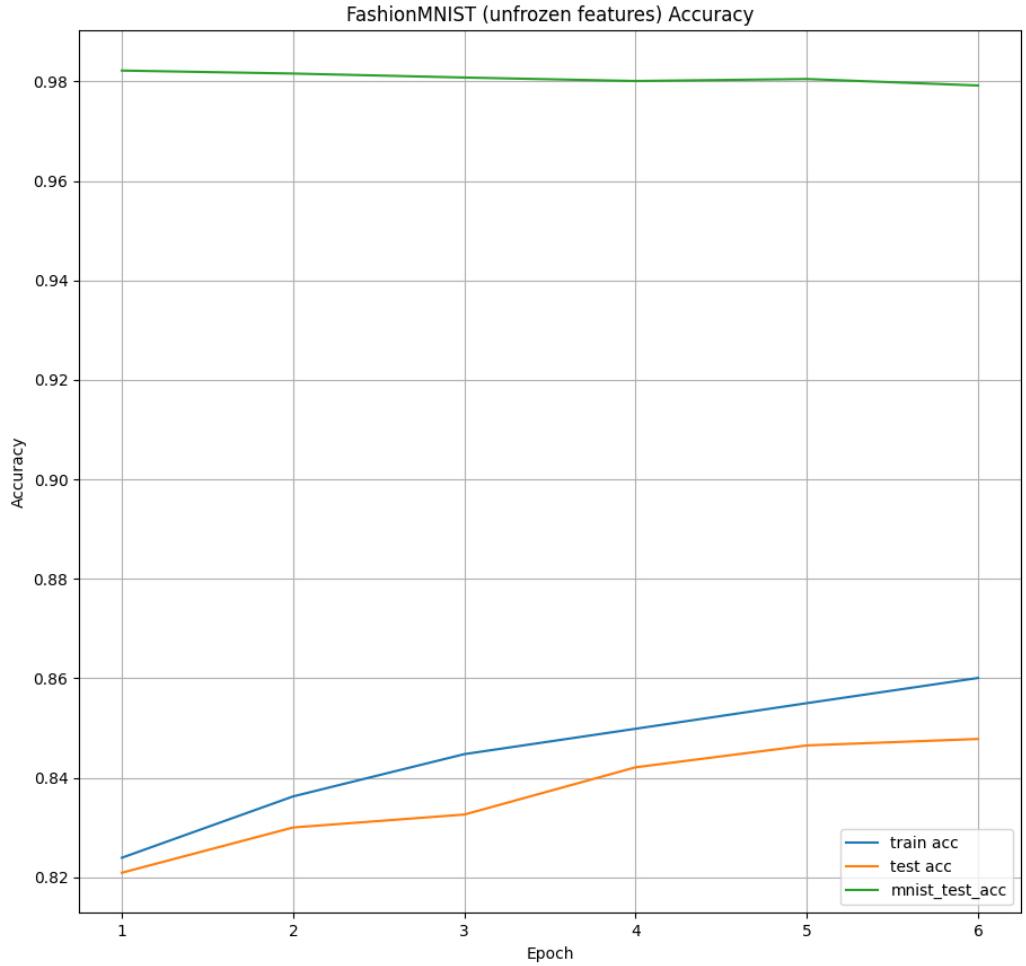

MNIST Loss
train loss
test loss
Loss
Epoch
1.0
0.8
0.6
0.4
0.2
1
2
3
4
5
6
7
8

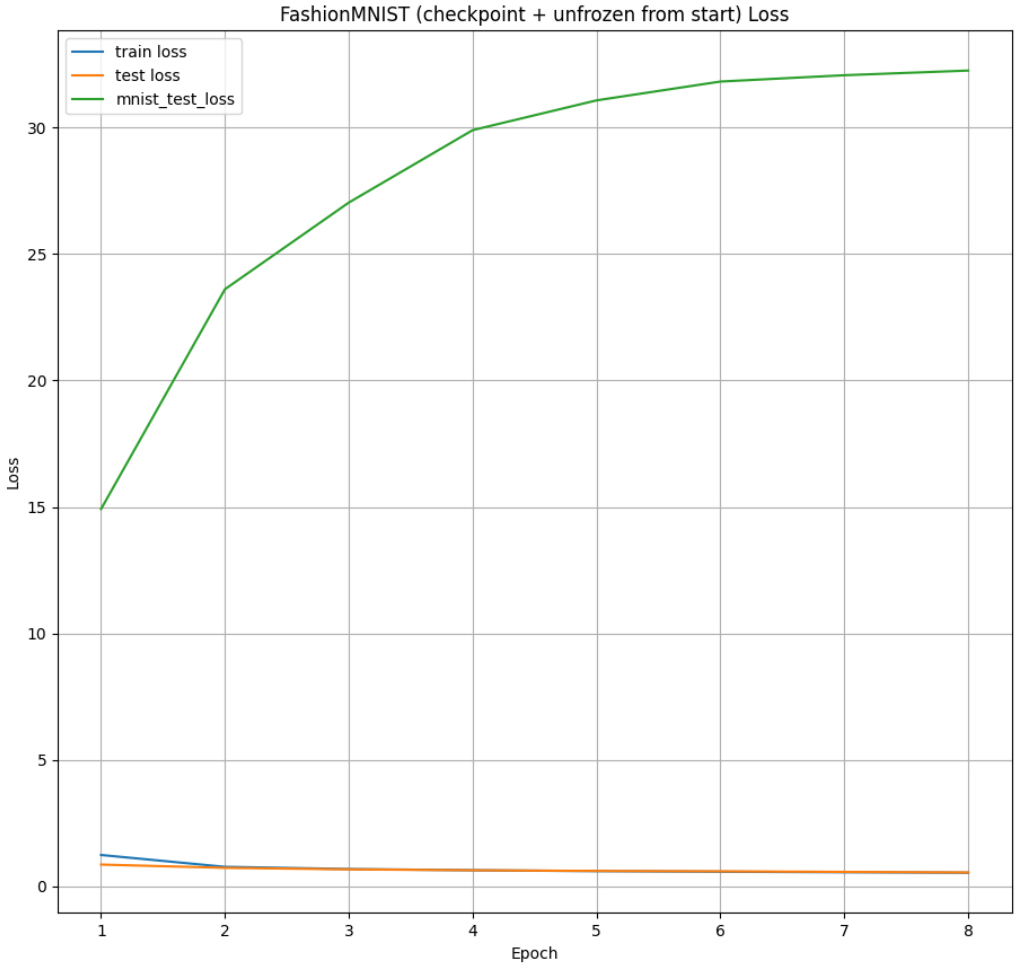

MNIST Accuracy
train acc
test acc
Accuracy
Epoch
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
1
2
3
4
5
6
7
8

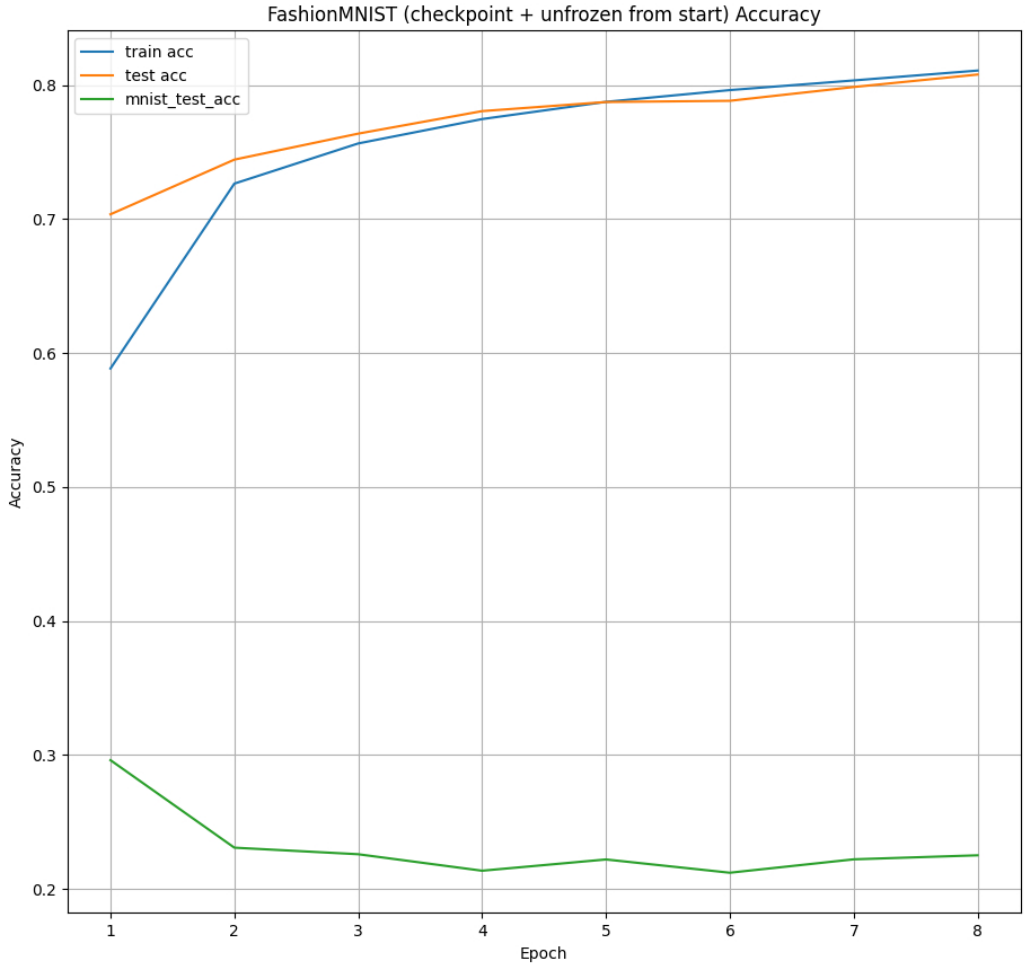

FashionMNIST (frozen features) Loss
train loss
test loss
Loss
Epoch
1.0
0.9
0.8
0.7
0.6
1
2
3
4
5
6
7
8

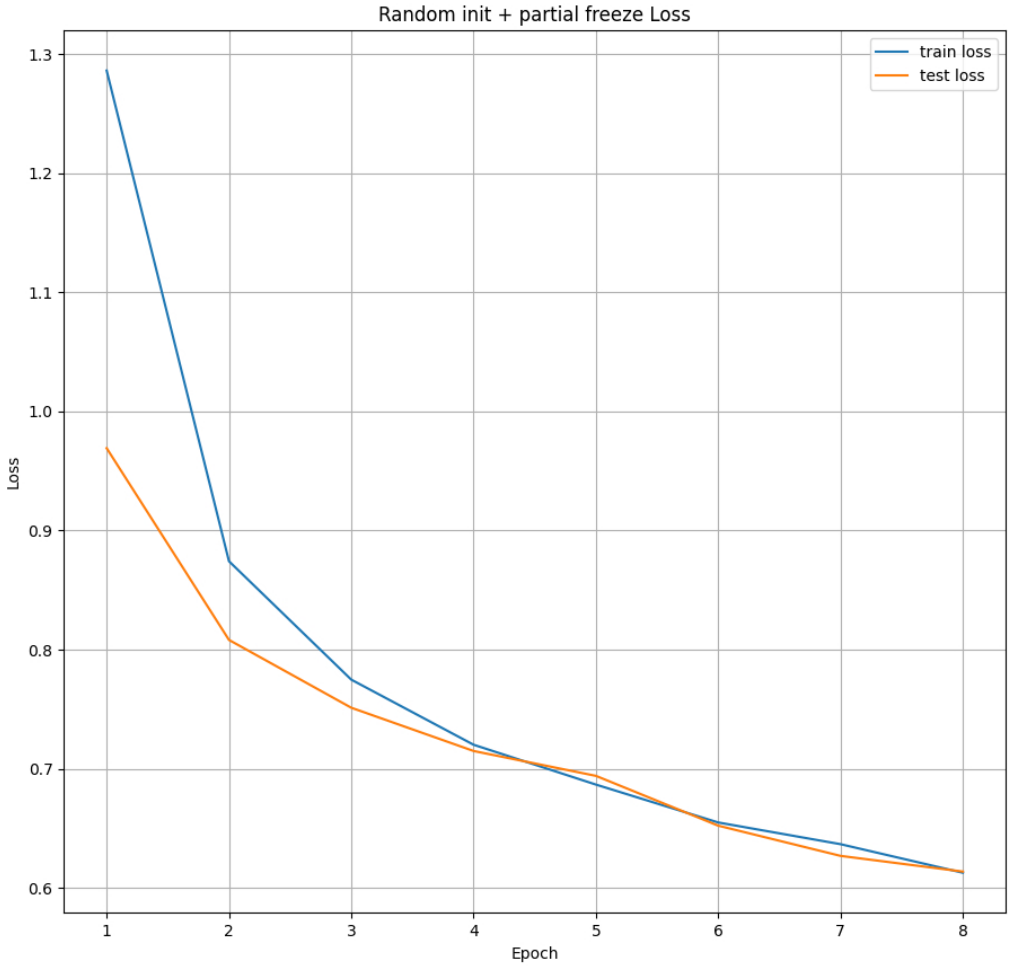

FashionMNIST (frozen features) Accuracy
train acc
test acc
Accuracy
Epoch
0.80
0.78
0.76
0.74
0.72
0.70
0.68
0.66
1
2
3
4
5
6
7
8

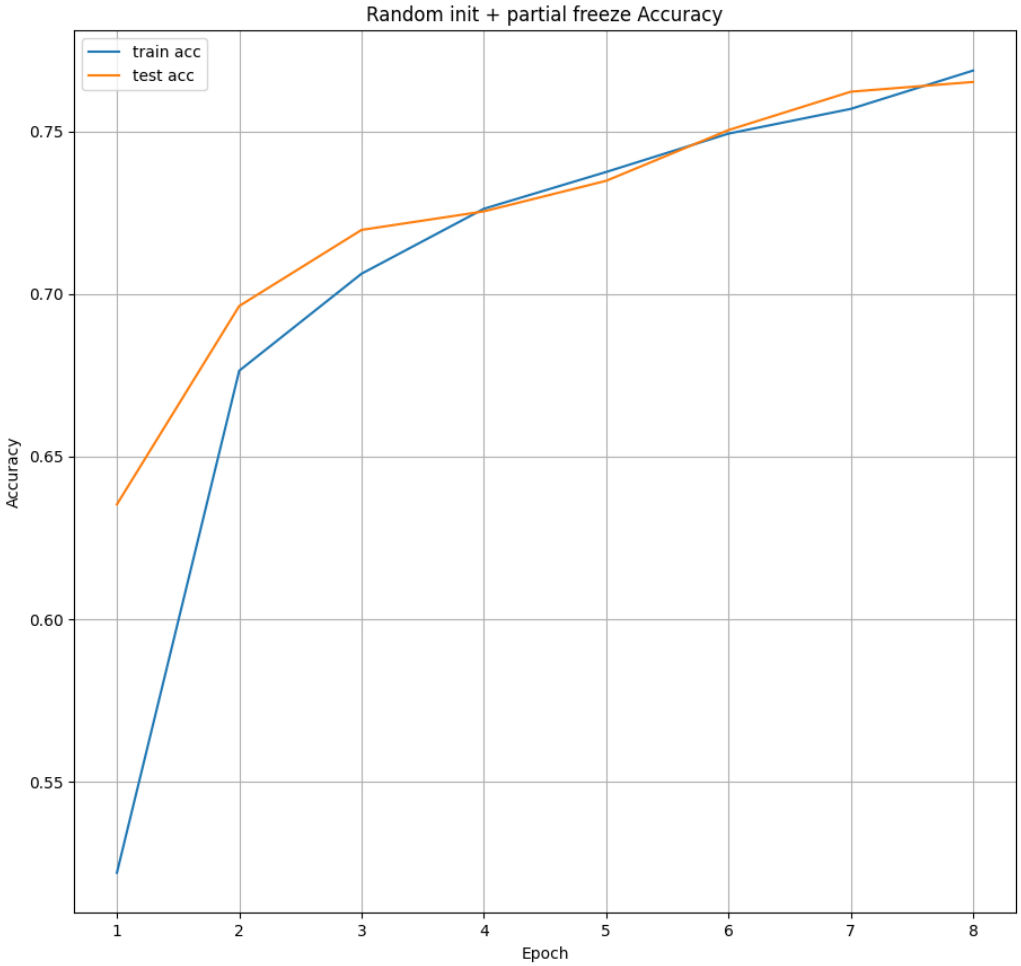

FashionMNIST (unfrozen features) Loss
train loss
test loss
mnist_test_loss
Loss
Epoch
0.5
0.4
0.3
0.2
0.1
1
2
3
4
5
6

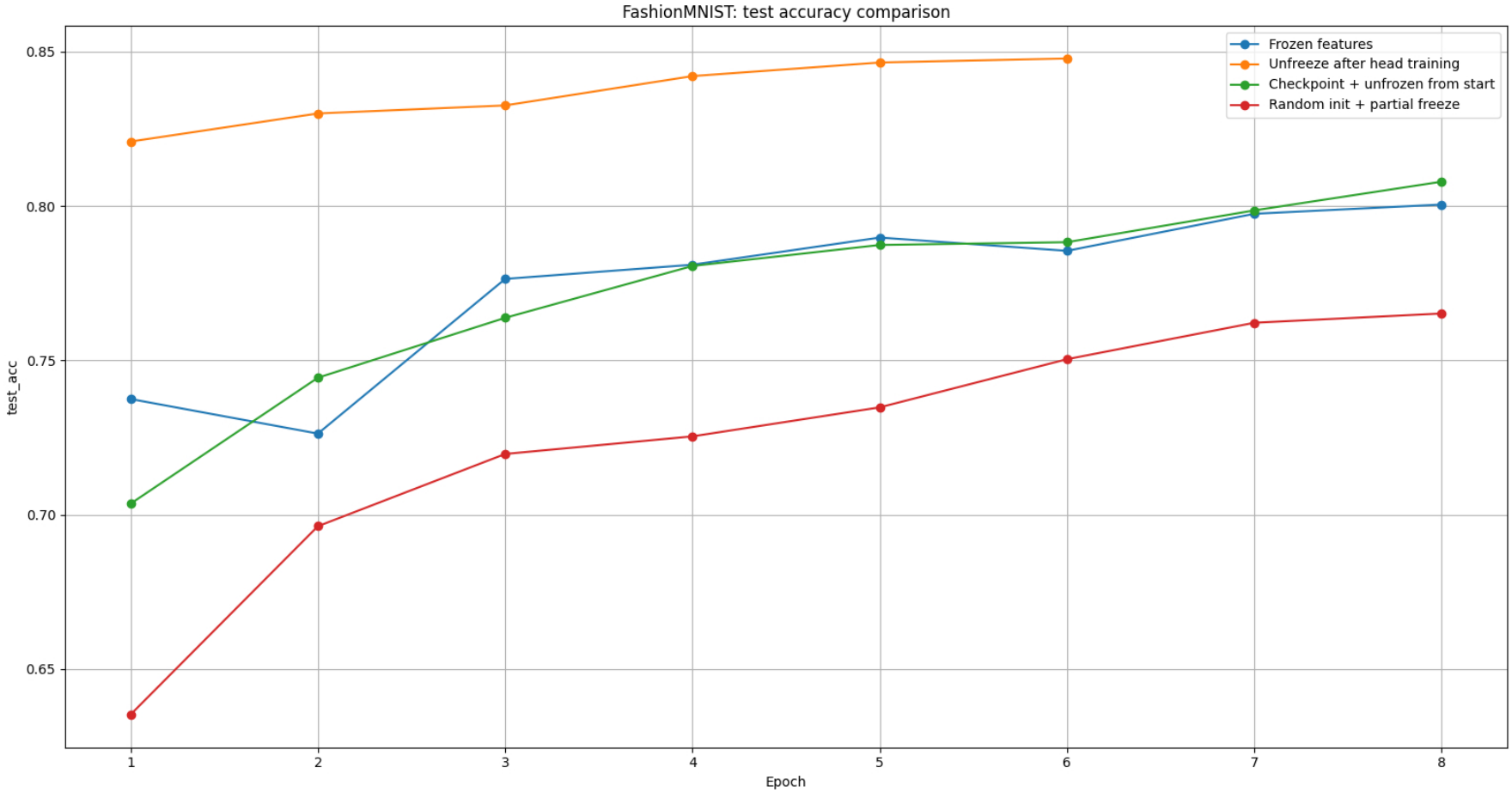

FashionMNIST (unfrozen features) Accuracy
train acc
test acc
mnist_test_acc
Accuracy
Epoch
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.84
0.82
1
2
3
4
5
6

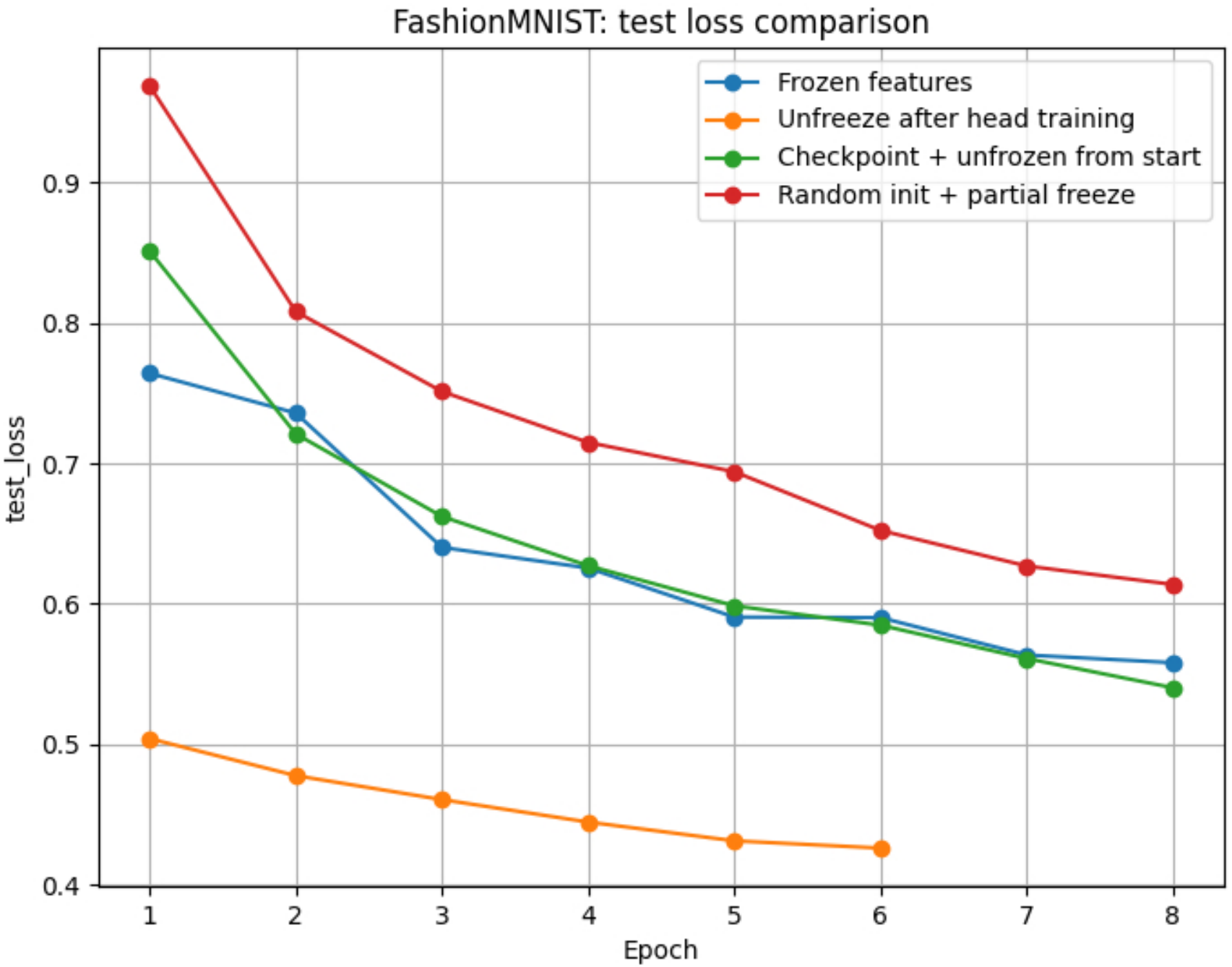

FashionMNIST (checkpoint + unfrozen from start) Loss
train loss
test loss
mnist_test_loss
Loss
Epoch
0
5
10
15
20
25
30
1
2
3
4
5
6
7
8

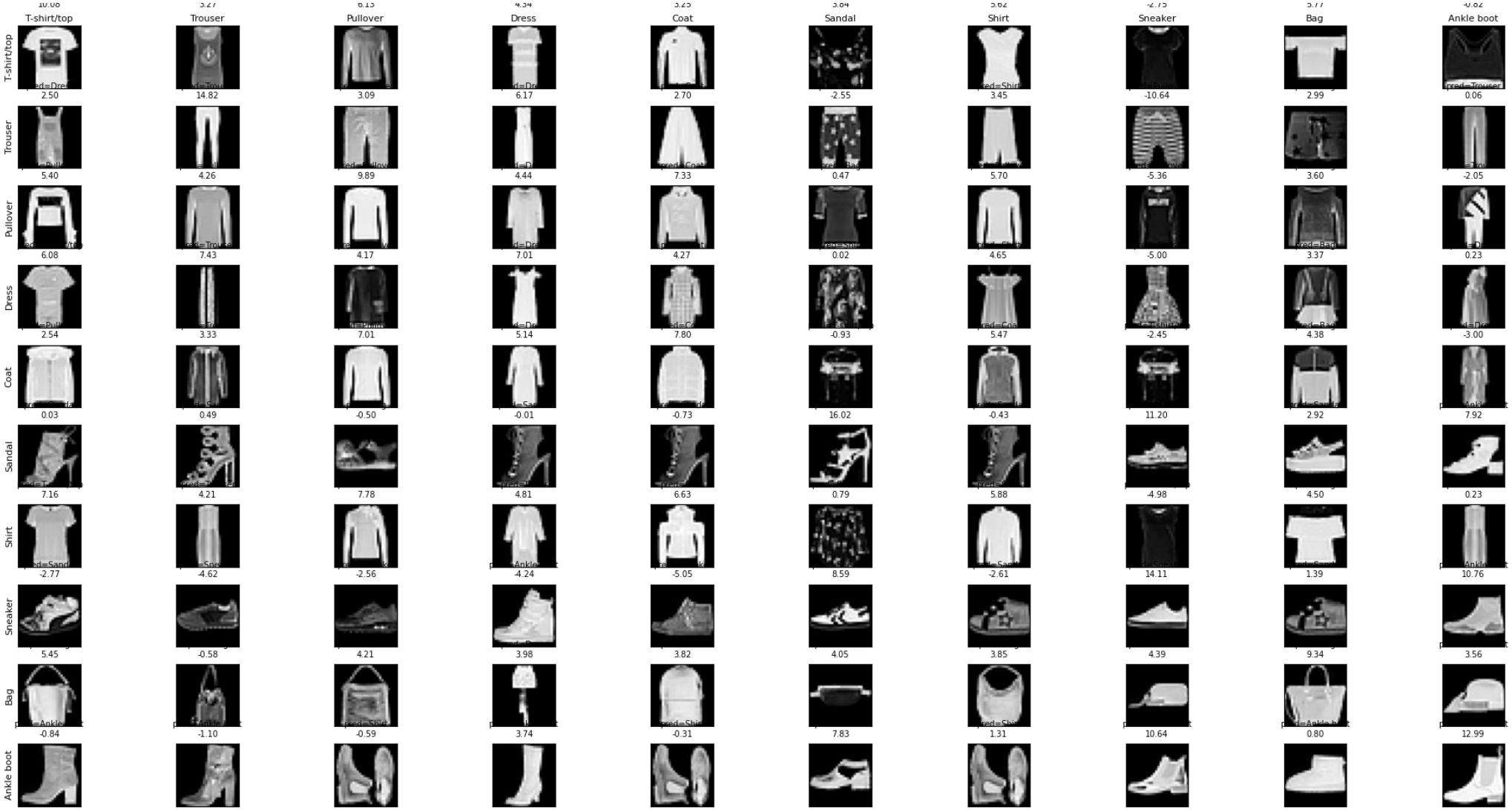

FashionMNIST (checkpoint + unfrozen from start) Accuracy
train acc
test acc
mnist_test_acc
Accuracy
Epoch
0.2
0.3
0.4
0.5
0.6
0.7
0.8
1
2
3
4
5
6
7
8


Random init + partial freeze Loss
train loss
test loss
Loss
Epoch
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
1
2
3
4
5
6
7
8


Random init + partial freeze Accuracy
train acc
test acc
Accuracy
Epoch
0.75
0.70
0.65
0.60
0.55
1
2
3
4
5
6
7
8


FashionMNIST: test accuracy comparison
Frozen features
Unfreeze after head training
Checkpoint + unfrozen from start
Random init + partial freeze
test_acc
0.85
0.80
0.75
0.70
0.65
Epoch
1
2
3
4
5
6
7
8


FashionMNIST: test loss comparison
Frozen features
Unfreeze after head training
Checkpoint + unfrozen from start
Random init + partial freeze
test_loss
0.9
0.8
0.7
0.6
0.5
0.4
1
2
3
4
5
6
7
8
Epoch


10.06
3.27
6.13
4.54
3.25
3.64
3.62
-2.75
5.77
-0.62
T-shirt/top
Trouser
Pullover
Dress
Coat
Sandal
Shirt
Sneaker
Bag
Ankle boot
T-shirt/top
2.50
14.82
3.09
6.17
2.70
-2.55
3.45
-10.64
2.99
0.06
Trouser
5.40
4.26
9.89
4.44
7.33
0.47
5.70
-5.36
3.60
-2.05
Pullover
6.08
7.43
4.17
7.01
4.27
0.02
4.65
-5.00
3.37
0.23
Dress
2.54
3.33
7.01
5.14
7.80
-0.93
5.47
-2.45
4.38
-3.00
Coat
0.03
0.49
-0.50
-0.01
-0.73
16.02
-0.43
11.20
2.92
7.92
Sandal
7.16
4.21
7.78
4.81
6.63
0.79
5.88
-4.98
4.50
0.23
Shirt
-2.77
-4.62
-2.56
-4.24
-5.05
8.59
-2.61
14.11
1.39
10.76
Sneaker
5.45
-0.58
4.21
3.98
3.82
4.05
3.85
4.39
9.34
3.56
Bag
-0.84
-1.10
-0.59
3.74
-0.31
7.83
1.31
10.64
0.80
12.99
Ankle boot